AI STUDY BUDDY API

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Project Overview

Objective: To develop an AI-based study assistant API.

The system helps students to:

- Ask study-related questions
- Get AI-generated answers
- Store study data
- Get personalized learning support

Tools & Technologies Used

Node.js – Backend runtime

Express.js – REST API development

MongoDB – Database

Gemini API – AI response generation

JavaScript – Programming language

Postman – API testing

Project Steps

1. Set up the Node.js environment
2. Create the Express.js server
3. Connect MongoDB database
4. Configure the Gemini API key
5. Create API endpoints
6. Test the API
7. Generate AI-based responses

Features

- Submit study questions
- Generate AI-based answers
- Store study data in MongoDB
- Manage study history
- REST API support
- Personalized study assistance

Benefits

- Helps students learn easily
- Provides quick answers
- Saves study time
- Stores study information
- Supports personalized learning
- Simple and interactive system

Output Demonstration

1. User sends a study question
2. API receives the question
3. Gemini API processes the question
4. AI response is generated
5. Study data can be stored in MongoDB

Conclusion

This project demonstrates the integration of Node.js, MongoDB and Gemini AI to create an intelligent study assistant API. It provides students with quick and useful learning support.

Future Enhancements

- Voice-based learning
- PDF and notes analysis
- Automatic quiz generation
- Study reminders
- Personalized learning plans
- Student progress tracking

THANK YOU

AI Study Buddy API

Smart Learning • Quick Answers • Personalized Support